NGUYEN NGOC DUC PHAT

Unity Gameplay Intern

☎ 0836663188 ✉ phatbo13579@gmail.com GitHub: github.com/KimgionDev Can Tho

OBJECTIVE

Software Engineering student at Can Tho University with experience in 2D game development using Unity and C#. Hands-on with gameplay systems, UI state flow, Utility AI and ScriptableObject-based data design through academic and personal projects. Seeking a Unity Gameplay Intern position at SEN CITY to work on gameplay systems and AI behavior in a real production team.

EDUCATION

Can Tho University | **Major:** Software Engineering | **GPA:** 3.56 / 4.0 2023 - 2027

Relevant Coursework: OOP, Data Structures & Algorithms, Software Architecture & Design, Database Management Systems.

Languages: English (basic, expected TOEIC 12/2026), Japanese (studying N5)

SKILLS

Game Engine & Languages	Unity (2D Core), C#, GAML (Simulation).
Gameplay Programming	Design Patterns (Singleton, Object Pooling, State Machine), Grid-based Systems.
Architecture & AI	MVC, MVVM, AI Behavior (Utility Matrix), BSP Algorithm, Agent-Based Modeling (GAMA Platform)
Tools & Platforms	Git, GitHub (PR workflow, CodeRabbit for automated review), AI-assisted coding (Antigravity, Codex)
Databases	SQL Server, MySQL.

GAME DEVELOPMENT PROJECTS

Mini Moving (Unity 2D - Mobile Puzzle Game) | *Team size: 3* Sep 2026 - Present

Gameplay & UI Developer

- **Description:** A 2D bin-packing puzzle game where players arrange cargo into a truck's grid within a "Time-Budget Economy" system.
- **Responsibilities:**
  - Developed core Gameplay UI (Time Gauge, Order Result) and created reusable Prefabs for dynamic screens (Job Cards, Day Summary).
  - Managed game State Flow by integrating UI systems smoothly into the main gameplay loop.
  - Built a modular Upgrade System using ScriptableObjects for data configuration, allowing easy balancing of player progression.
  - Wrote Automated Tests to validate core gameplay logic and upgrade mechanics, ensuring system stability.
- Source: Atus229/CT239_ProjectGame

Apocalypse Grid (Unity 2D - Top-Down Shooter) | *Personal Project* May 2026 - Aug 2026

Solo Developer

- **Description:** A Rogue-lite 2D Top-Down Shooter focusing on infinite replayability through procedural generation.
- **Responsibilities**
  - Generated dungeons with BSP (loop and Queue instead of recursion) and shaped walls with Random Walk and bitmasking.
  - Added object pooling for bullets and enemies; runs at 60+ FPS with hundreds of enemies per level (about 300 at level 50).
  - Wrote enemy AI (chasing, obstacle avoidance, wave respawn) and 360° aiming and shooting.
  - Built the gameplay loop (loot drops, quest items that unlock the next level) and an upgrade dashboard with ScriptableObjects (health, damage, speed, ammo).
- Source: KimgionDev/Apocalypse-Grid | Demo: Play in browser

Fire Evacuation Simulation (GAMA Platform) | *Team size: 3* Sep 2026 - Present

Team Leader & Designer

- **Description:** An Agent-Based Modeling (ABM) simulation of emergency evacuation during a fire incident.
- **Responsibilities:**
  - Led a 3-person team: wrote the design document and 5-week plan, and designed the Utility AI logic (action scoring, penalties, priority bonuses, tie-breaking) for evacuees, firefighters and rescuers.
  - Built the 50x50 office map in GAML and a navigation graph so agents path-find around walls and obstacles.
  - Built real-time charts for escaped evacuees, casualties and panic levels.

- Source: KimgionDev/fire-evac-sim

OTHER EXPERIENCES

CTU-Scheduler (University Research Project) | *Team size: 4* Apr 2026 - Oct 2026

Member

- Built Excel export (including multiple timetables at once) and a timeline view of class data fetched from the university website.
- Wrote project documentation.
- Source: d3nhatv0lam/CTU-Scheduler

VocabMaster (Desktop Application) | *Course Project* Jan 2026 - Mar 2026

Solo Developer

- Built a vocabulary app in C# WinForms with SQL Server (ADO.NET): auto translation and dictionary lookup, flashcards, and a quiz that collects wrong answers into a “hard words” list.
- Added a dashboard with pie and line charts for topic distribution and quiz scores over time, and JSON import/export for backup.
- Source: KimgionDev/VocabMaster-Winform